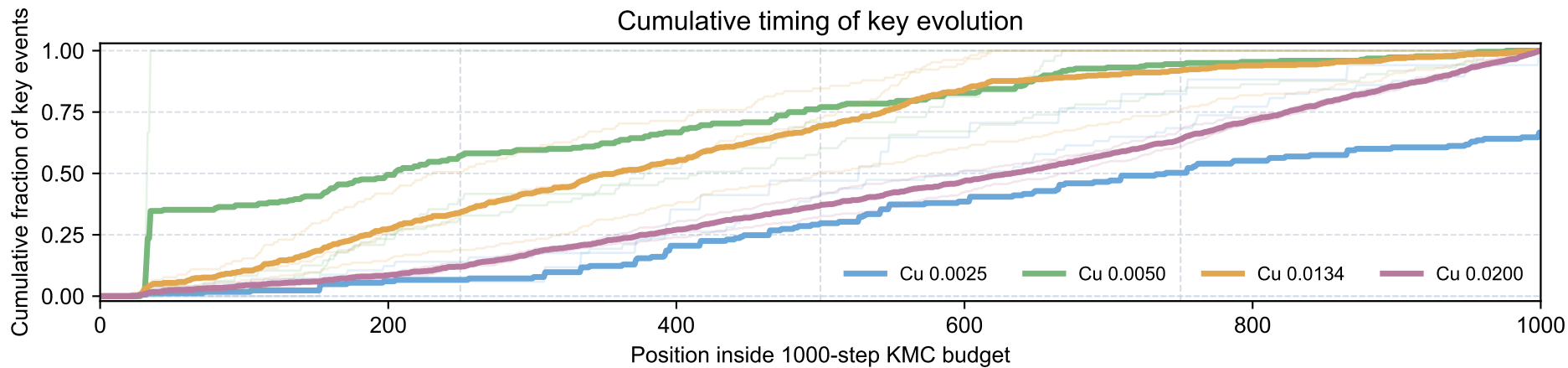


Random timing of key structural evolution under a fixed 1000-step KMC budget

Event raster: exact positions of key evolution within the 1000-step budget



Cumulative timing of key evolution



Diagnostic proxy: key evolution = Cu-vacancy exchange event. Each row is one teacher trajectory; faint curves are seeds, bold curves are density means.